

Verity: Graph-Based Source Credibility Inference for Autonomous Information Networks

Cole Hoke

Abstract

In this paper, we present Verity, a graph-based framework for inferring source credibility from the structure of an information network. The information network is represented as a bipartite graph, where nodes correspond to sources and claims, and edges correspond to assertions linking sources to the claims they assert. Source credibility is estimated by applying an iterative propagation algorithm in which credibility is repeatedly propagated between sources and claims until a stable credibility distribution is reached. Unique properties of the proposed approach are explored in several experiments, which include convergence analysis, the initial allocation of credibility, graph connectivity effects, claim representation, and the incorporation of agreement-weighted propagation. It is shown that credibility can be estimated directly from the graph structure while allowing additional structural information to be incorporated into the propagation process. We believe that graph-based credibility inference provides a promising foundation for credibility estimation within the emerging Agentic Web.

1 Introduction

In modern information systems, knowledge is gathered from vast quantities of various sources. These systems form the basis of many tools that we use today, ranging from search engines and knowledge bases to recommendation systems and scientific databases. One of the largest repositories of information used by these systems is the web. Today, the web is experiencing a fundamental shift from human-to-human to machine-to-machine information exchange.

With advancements in artificial intelligence systems, particularly large language models (LLMs) [6] and autonomous agents [7], this transition has accelerated rapidly. Information is now being generated, distributed, and

consumed with minimal or no human intervention. Autonomous agents routinely retrieve information from multiple sources and use it to make decisions and execute downstream tasks across digital environments [5]. Hence, the credibility of this information is critical for the actions that follow. Existing approaches to validate such information typically rely on external evaluation such as human review or verification against trusted sources. More recently, information is often routed through a secondary large language model for evaluation. These approaches can be effective, but constantly relying on external evaluation for every decision can be slow and computationally expensive while becoming increasingly difficult to integrate as autonomous agents continue to operate at greater pace and scale. This is because credibility cannot be computed within the information network itself, but instead must be established through external evaluation.

The difficulty is not just the fact that different sources disagree. Instead, information often tends to propagate from source to source through various ways such as copying or syndication. This means that seeming agreement between several sources is not necessarily independent evidence. Ten sources repeating the same wrong claim may provide less evidence than two sources agreeing independently.

To bridge this structural gap, we introduce Verity, a graph-based framework for inferring source credibility in large collections of conflicting claims. It estimates source credibility from the network of assertions connecting sources and claims. These credibility estimates allow autonomous systems to distinguish between conflicting information without external evaluation. This paper introduces the Verity framework, describes the credibility inference algorithm, and discusses its potential role in information exchange within the emerging Agentic Web ecosystem. “tex

2 Verity Framework

The Verity credibility graph is an information model based on the network structure consisting of sources and claims. Let us define it as:

$$G = (S, C, E)$$

where S is the set of sources, C is the set of claims, and E is the set of assertions which link sources to the claims they assert. This approach allows us to represent the information distribution across independent publishers as a graph. We use this graph as the foundation for credibility inference throughout the framework.

2.1 Sources

A source refers to an individual publisher that asserts one or more claims. Examples of sources may include websites, organizations, and databases. Every source is mapped to a unique node in the credibility graph.

2.2 Claims

A claim refers to a piece of information about an underlying entity. The same claim may be asserted by multiple different sources. Whereas other sources may assert conflicting claims about that same entity. Claims form nodes that connect different sources within the credibility graph.

2.3 Assertions

Assertions refer to the relationship that exists between a source and a claim. Whenever a source makes a claim, an assertion is created which connects the corresponding source node to the claim node. Assertions are what form the overall graph structure, revealing the distribution of claims amongst several sources.

2.4 Credibility Graph

The credibility graph is made up of two types of nodes: sources and claims. The assertions link these node types to form a bipartite graph where sources can only connect with claims and claims can only connect with sources. Figure 1 pictures a simplified example of the graph.

“ “latex

3 Credibility Inference

The credibility graph defines how sources and claims are connected, but does not, on its own, provide any information about the credibility of either. Next, we describe an iterative inference algorithm that computes the credibility of each source, based on the properties of the network.

In this way, the algorithm keeps iterating between the source nodes and the claim nodes. Sources distribute their credibility among the claims they assert, and then these claims redistribute their accumulated support back to the sources that assert them. Repeating this process causes the source credibility vector to converge toward a stable distribution.

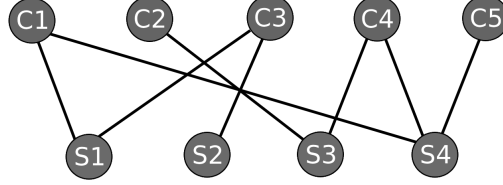


Figure 1: Source nodes (bottom) connect to the claim nodes (top) they assert. Each edge indicates one assertion.

3.1 Propagation Algorithm

Propagation consists of four stages that are done at each iteration step. In the first one, source credibility is propagated to the claims they assert. After this stage, in the second one, support for claims is aggregated from all asserting sources. After the aggregation process, the result is propagated back to the source nodes. Finally, the credibility vector is normalized before the next iteration begins. The following subsections describe each stage formally.

3.2 Claim Support

For the first stage in each iteration, the support for each claim is computed. In doing so, each of the sources distributes their credibility evenly across all of the claims that it asserts. Hence, sources that assert a lot of claims will have less credibility per claim as compared to the sources that assert fewer claims.

Claim c_j receives support according to the formula:

$$c_j = \sum_{i \in A(j)} \frac{s_i}{d_i}.$$

Here, s_i is the credibility of source i , d_i is the number of claims from source i , and $A(j)$ is the set of sources asserting claim j .

3.3 Source Update

The second step of the iteration process is updating the credibility of each source. This is done using the support from each claim that the source asserts. Because claims can be asserted by multiple sources, the support for each claim is divided evenly amongst all of the sources asserting it. Therefore, a source with highly supported claims will have high credibility.

The new credibility of source s_i is calculated according to the formula:

$$s'_i = \sum_{j \in C(i)} \frac{c_j}{a_j}. \quad (1)$$

Here, c_j is the support of claim j , a_j is the number of sources asserting claim j and $C(i)$ is the set of claims asserted by source i .

3.4 Normalization

The obtained credibility vector must be normalized after each step before moving on to the next step to maintain proportionality of the sources' credibility without changing its norm in future iterations.

We normalize the credibility of source s_i according to the following formula:

$$s_i = \frac{s'_i}{\sum_k s'_k}. \quad (2)$$

where s'_i is the updated credibility of source i before normalization.

3.5 Convergence

The propagation process continues repeatedly until the source credibility vector converges to a stable distribution. In each step, the obtained source credibility vector is compared with the source credibility vector from the previous step. The convergence process is completed when the maximal difference between corresponding credibility values falls below a predefined threshold.

Convergence criterion:

$$\max_i \left| s_i^{(t+1)} - s_i^{(t)} \right| < \varepsilon, \quad (3)$$

where $s_i^{(t)}$ is the credibility of source i after iteration t , and ε is the threshold value of convergence.

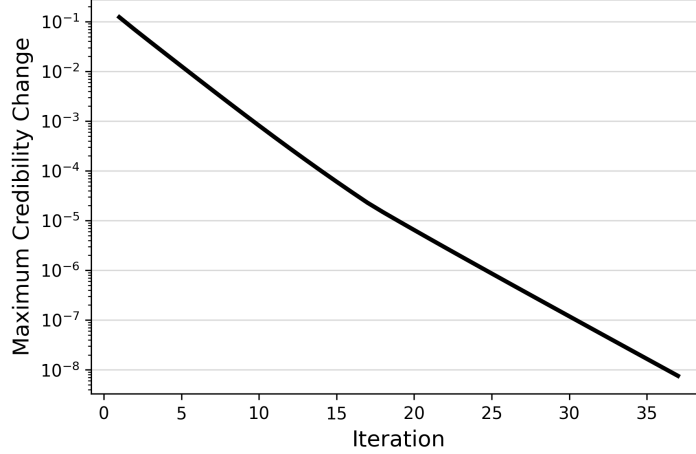


Figure 2: Maximum change between consecutive source credibility vectors during propagation. The credibility vector converges after 37 iterations.

4 Experimental Evaluation

The proposed algorithm is studied by performing several experiments to explore its properties such as convergence, robustness, and behavior under various graph structures. Unless stated otherwise, all experiments have been performed on the Verity credibility graph presented in Section 2.

4.1 Convergence Behavior

First, we study the convergence properties of the proposed propagation algorithm to determine whether it converges to a stable credibility distribution. In order to achieve this, we initialize all sources with equal starting credibility and apply the propagation algorithm described in Section 3 iteratively until the convergence condition is met. We measure the maximum absolute difference between the source credibility vectors of two consecutive iterations during the propagation process.

We can see in Figure 2 that the maximum difference in source credibility values diminishes at each subsequent iteration up to the point where the convergence condition is met.

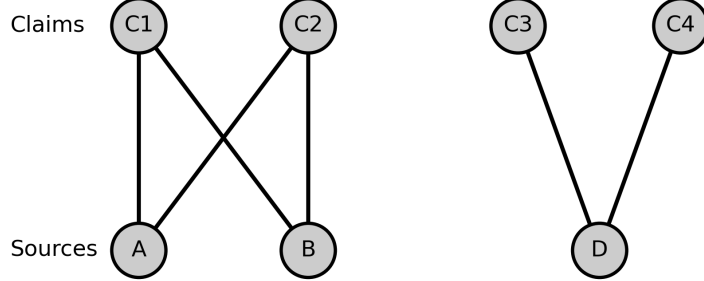


Figure 3: Toy credibility graph consisting of two connected components. Sources A and B share claims $C1$ and $C2$, while source D is isolated through claims $C3$ and $C4$. Credibility propagates only within connected components.

4.2 Initialization Independence

Having demonstrated convergence of the propagation algorithm, we now explore whether the resulting credibility distribution is independent of the initial distribution. To perform this analysis, we test two methods of initialization. The first method assigns each source equal initial credibility. The second method gives each source a random credibility value, which is then normalized.

We observe that, despite starting at different points, both methods lead to the same credibility distribution. This suggests that the converged solution may be independent of the initial distribution. The resulting credibility distribution is therefore stable and consistent.

4.3 Graph Connectivity

Having explored the convergence properties of the propagation algorithm, we now turn our attention to the effect that graph connectivity has on credibility propagation. Since the process of credibility propagation occurs alongside assertion edges between source nodes and their claims, graph connectivity impacts which sources can propagate their credibility to one another.

For the purpose of studying the dynamics of this phenomenon, we first consider the toy graph shown in Figure 3.

It turns out that credibility propagates only among the connected com-

ponents of the graph. Sources disconnected from the rest of the network do not receive credibility from outside and thus converge independently of the rest of the graph.

This example clearly shows that graph connectivity plays an important part in credibility inference. The degree to which the sources share claims determines the pattern of credibility propagation across the network.

4.4 Claim Representation

In view of the above observations regarding the effect of graph connectivity on credibility propagation, let's examine the effects that the choice of representation of claims has on the graph structure. As the credibility graph is constructed from the set of claims and assertions, the way claims are represented affects the graph structure in a significant way.

In particular, initially, claims were considered as unique combinations made up of an entity, its attribute, and the observed value of the attribute. While such a representation kept track of all observed values, it led to a highly disconnected graph in which many claims were asserted by only one source. Therefore, the credibility propagation was restricted to small connected components of the graph.

To address this problem, we redefine a claim as an ordered pair of the entity and attribute alone. Using this definition, multiple sources asserting different values for the same attribute get connected through a common claim node instead of being kept separate as distinct nodes.

We find this way of representing claims causes a significant increase in the connectivity of the graph. Credibility can now spread through a larger part of the graph. Consequently, conflicting assertions form a single inference path, thus allowing us to infer the credibility from the graph structure as a whole.

4.5 Agreement-Weighted Propagation

After improving the connection of our credibility graph, we explore whether the propagation algorithm can exploit the level of agreement between contradicting assertions. In the original propagation algorithm, all the assertions carried equal influence during the credibility propagation regardless of the popularity of the asserted value.

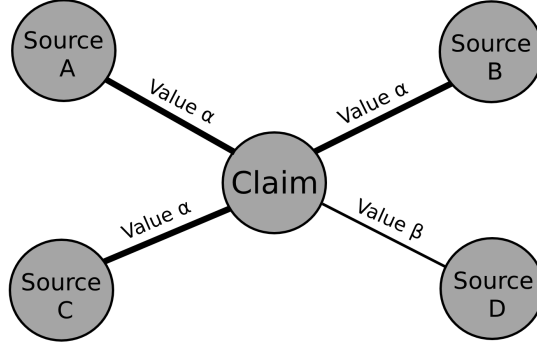


Figure 4: Agreement-weighted propagation. Four sources assert competing values for the same claim. Assertions supporting the most frequently asserted value receive greater propagation weight than assertions supporting less common values.

This is where agreement-weighting comes into play. The assertions on each claim are weighted depending on the percentage of the sources asserting that particular value. Assertions which support more popular values are given higher weights compared to those which assert rare values, as pictured in Figure 4.

We have seen that using agreement weighting enables us to incorporate local agreement information into the propagation process while retaining the convergence properties of the algorithm. As opposed to the original process where all the assertions were treated equally, we are now considering the distribution of the competing values on the same claim.

5 Discussion

The experiments performed in this paper illustrate several properties of the proposed framework. First of all, the propagation algorithm has been shown to stabilize at some credibility distribution and this distribution does not depend on the way sources are initialized. Consequently, one could reasonably assume that credibility estimates depend on the credibility graph structure and not on the initial credibility allocation.

Another property worth noting is the role of graph connectivity. Disconnected sources evolve independently; therefore, their credibility estimates do not propagate through the whole network. It is clear that this considera-

tion called for a new approach to modeling claims. In the experiments, the definition of claims changed to entities and attributes instead of single value which increased graph connectivity and allowed conflicting assertions to be treated within the same inference process.

Furthermore, agreement-weighted propagation showed that the propagation algorithm allows for more information to be used during the propagation process without any additional evaluation. The propagation algorithm can use the level of local agreement between conflicting values as the source of information.

Nonetheless, there are still some problems related to the framework that deserve further research. Graph-based credibility inference shares similarities with other iterative graph ranking methods such as PageRank [1] and HITS [3], as well as truth discovery methods including TruthFinder [8] and subsequent surveys of truth discovery [4]. However, one of the main problems in credibility inference is distinguishing genuine independent evidence from repeated information. The fact that many sources agree upon something does not automatically mean that the consensus is independent because information may propagate due to copying and syndication. Hence, the agreement overestimates the support for a particular claim considerably. Explicitly modeling source dependencies is one of the most important directions for future graph-based credibility inference [2].

6 Conclusion

The emergence of the Agentic Web creates a growing need for scalable credibility inference techniques which will not depend on constant external evaluation. In this paper, we have proposed the Verity method for calculating source credibility as the property of an information network in the form of a graph. Instead of relying on trusted reference data or human verification, Verity estimates the credibility directly from the relationships between sources and claims.

Our iterative propagation approach propagates credibility between the source and claim nodes until a stable credibility distribution is reached. Through a series of experiments, we have shown that the proposed algorithm is consistent and stable, that the credibility distribution does not depend on the chosen initialization and that the graph connectivity plays an important role in the credibility inference process. Moreover, we have proven that the improvement of the graph connectivity through the proper representation of claims and agreement-weighted propagation provides additional structural

information that is useful for credibility inference.

We believe that credibility inference is inherently a graph problem. Through a representation of sources, claims, and assertions, Verity provides a framework through which we calculate credibility based on the properties of the graph directly from the structure of the information network. We hope this will lay the groundwork for credible inference as a core capability of future autonomous information systems.

6.1 Future Work

While the proposed framework provides the groundwork for graph-based credibility inference, there is still much research to be done. Specifically, agreement among several sources does not automatically mean independence, since information may propagate among sources. Thus, explicitly modeling source dependencies becomes an important direction for future work.

Graph-based credibility inference will be further developed as autonomous information systems keep improving and become more dependent on information collected from multiple independent sources.

References

- [1] Sergey Brin and Lawrence Page. The anatomy of a large-scale hypertextual web search engine. *Computer Networks and ISDN Systems*, 30(1–7):107–117, 1998.
- [2] Xin Luna Dong, Laure Berti-Equille, and Divesh Srivastava. Integrating conflicting data: The role of source dependence. *Proceedings of the VLDB Endowment*, 2(1):179–190, 2009.
- [3] Jon M. Kleinberg. Authoritative sources in a hyperlinked environment. *Journal of the ACM*, 46(5):604–632, 1999.
- [4] Yaliang Li, Jing Gao, Chuishi Meng, Qi Li, and Jiawei Han. A survey on truth discovery. *ACM SIGKDD Explorations Newsletter*, 17(2):1–16, 2016.
- [5] Tula Masterman, Sandi Besen, Mason Sawtell, and Alex Chao. The landscape of emerging ai agent architectures for reasoning, planning, and tool calling: A survey. *arXiv preprint arXiv:2404.11584*, 2024.
- [6] OpenAI. Gpt-4 technical report. *arXiv preprint arXiv:2303.08774*, 2023.

- [7] Zhiheng Xi, Wenxiang Chen, Xin Guo, Wei He, Yi Ding, Yongting Chai, et al. The rise and potential of large language model based agents: A survey. *arXiv preprint arXiv:2309.07864*, 2023.
- [8] Xiaoxin Yin, Jiawei Han, and Philip S. Yu. Truth discovery with multiple conflicting information providers on the web. *IEEE Transactions on Knowledge and Data Engineering*, 20(6):796–808, 2008.